COURSE STRUCTURE:

Module_1: Introduction to Artificial Intelligence and Soft Computing

Introduction and definition of Artificial Intelligence, Intelligent Agents, Agents and Environment, Rationality, Nature of environment, Structure of Agents, Types of agents (Simple Reflex, Model-Based, Goal-Based, Utility-Based).

Soft computing: Introduction to Soft Computing, meaning and concept of soft computing, soft computing vs hard computing, Handling uncertainty, imprecision, and ambiguity, Limitations of conventional (hard) computing, Components of Soft Computing. Some application of soft computing techniques.

Module-2: Problem solving be Intelligent Search

Uninformed (Blind) Search Strategies Breadth-First Search (BFS), Depth-First Search (DFS),

Search with partial information (Heuristic or Informed Search) - Hill climbing (simple hill climbing, steepest Ascent), A* and AO* (AND-OR search) algorithm.

Module-3: Artificial Neural Network

Artificial Neural Network Architecture**,** Single layer and multi layer feed forward networks, Learning in Neural Networks -Supervised learning – Delta learning rule. Backpropagation algorithm.

Module-4: Fuzzy Logic

Introduction to Fuzzy set: Fuzzy set theory, fuzzy vs crisp set, membership function and types of membership function. Fuzzy logic: Fuzzy logic basics, fuzzy rules and reasoning, fuzzy inference system- Sugeno and Mamdani based Fuzzy Inference Systems, Defuzzification methods and fuzzy controllers. Adaptive Neuro-Fuzzy Inference System (ANFIS): ANFIS architecture and ANFIS based on the Sugeno Fuzzy Inference System.

Module-5: Evolutionary algorithms

Biological background, Traditional optimization and search techniques, Definition of Genetic Algorithm, Genetic algorithm vs traditional algorithms, Components of Genetic Algorithm, Working of Genetic Algorithm, Selection, Crossover, and mutation methods, simple genetic algorithm, operators in genetic algorithm, stopping condition of Genetic Algorithm, constraints of Genetic Algorithm, problem solving using Genetic Algorithm,

the Schema theorem, Application of Genetic Algorithm, Advantages and limitations of Genetic Algorithm.

Particle Swarm Optimization: Basic Concepts (particle, swarm, search space, fitness function, position), Working of PSO (Initialization of particles, Fitness evaluation, Personal Best (pBest), Global Best (gBest), Updating velocity, Updating position, Stopping criterion). Neuro-fuzzy systems with metaheuristic optimization techniques like PSO.